A Major Project Presentation
on
Space Weather Prediction Platform (CosmoPredict)

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Content

Introduction

CosmoPredict: 3D Space Weather Forecast & Rocket Simulation

CosmoPredict

- Predicts solar flares and geomagnetic storms using Machine Learning.
- Combines historical and real-time space data from NASA, NOAA, and ACE satellites.
- Provides a futuristic **interactive dashboard** with 3D rocket trajectory simulation.
- Enhances safety for satellites, astronauts, and space missions.

Problem Statement

Space weather events like solar flares and geomagnetic storms pose serious risks to satellites, astronauts, and communication systems. Current monitoring systems collect vast data but lack predictive accuracy and real-time alerts. **CosmoPredict** addresses this gap by integrating AI and machine learning to forecast space weather events and provide actionable insights for timely preventive measures.

Unpredictable Events

Space weather events are **unpredictable**, affecting satellites, GPS, aviation, and power grids.

No Centralized Platform

Lack of **centralized, real-time monitoring platforms**.

Missing Alerts

Absence of **early warning alerts** for astronauts and space agencies.

Limited Guidance

Limited **operational guidance** for satellite trajectory adjustments during severe events.

Objectives

ML-Powered Predictions

Predict future solar flares and geomagnetic storms using **LSTM/GRU ML models**.

Real-Time Monitoring

Provide **real-time monitoring and interactive visualizations** on a web dashboard.

Automated Alerts

Generate **automated alerts** for high-risk events (dashboard, email, SMS).

Route Diversion

Introduce a **Route Diversion Module** to suggest safe orbital paths.

Educational Platform

Deliver an **educational and research-friendly platform** for space weather study.

Our AI-Driven Solution

Predictive AI

LSTM/GRU models forecast solar activity

Automated Alerts

Email, SMS, and dashboard warnings

Real-Time Monitor

Interactive dashboard with live feeds

Route Diversion

Suggest safe orbital adjustments

System Features

Real-Time Monitoring

Real-time space weather monitoring.

Predictive Analytics

Predictive analytics with historical data comparison.

3D Simulation

3D interactive rocket simulation for multiple deviation paths.

Alert System

Alert notifications via dashboard, email, and SMS.

Data Access

Historical data query and CSV/JSON download.

Responsive UI

Responsive UI with **dark/light mode toggle**.

Tools & Technologies

Frontend

React JS, Three.js, Tailwind CSS, Plotly/Chart.js.

Backend

Django REST Framework, PostgreSQL database.

ML Frameworks

TensorFlow, PyTorch.

Data Sources

NASA OMNIWeb, NOAA SWPC, ACE Satellite Data.

Others

GitHub for version control, cloud servers for hosting (AWS/GCP/Render).

System Architecture

A comprehensive workflow from data collection to actionable insights.

Data Collection

Fetch historical & real-time data from NASA, NOAA, ACE.

ML Prediction Module

LSTM/GRU models for solar flare & storm forecasting.

Web Dashboard

Visualization, alerts, historical comparison.

Route Diversion Module

Suggest alternative orbital paths for satellites.

Validation

Compare predictions against known historical events.

ER Diagram

Entities

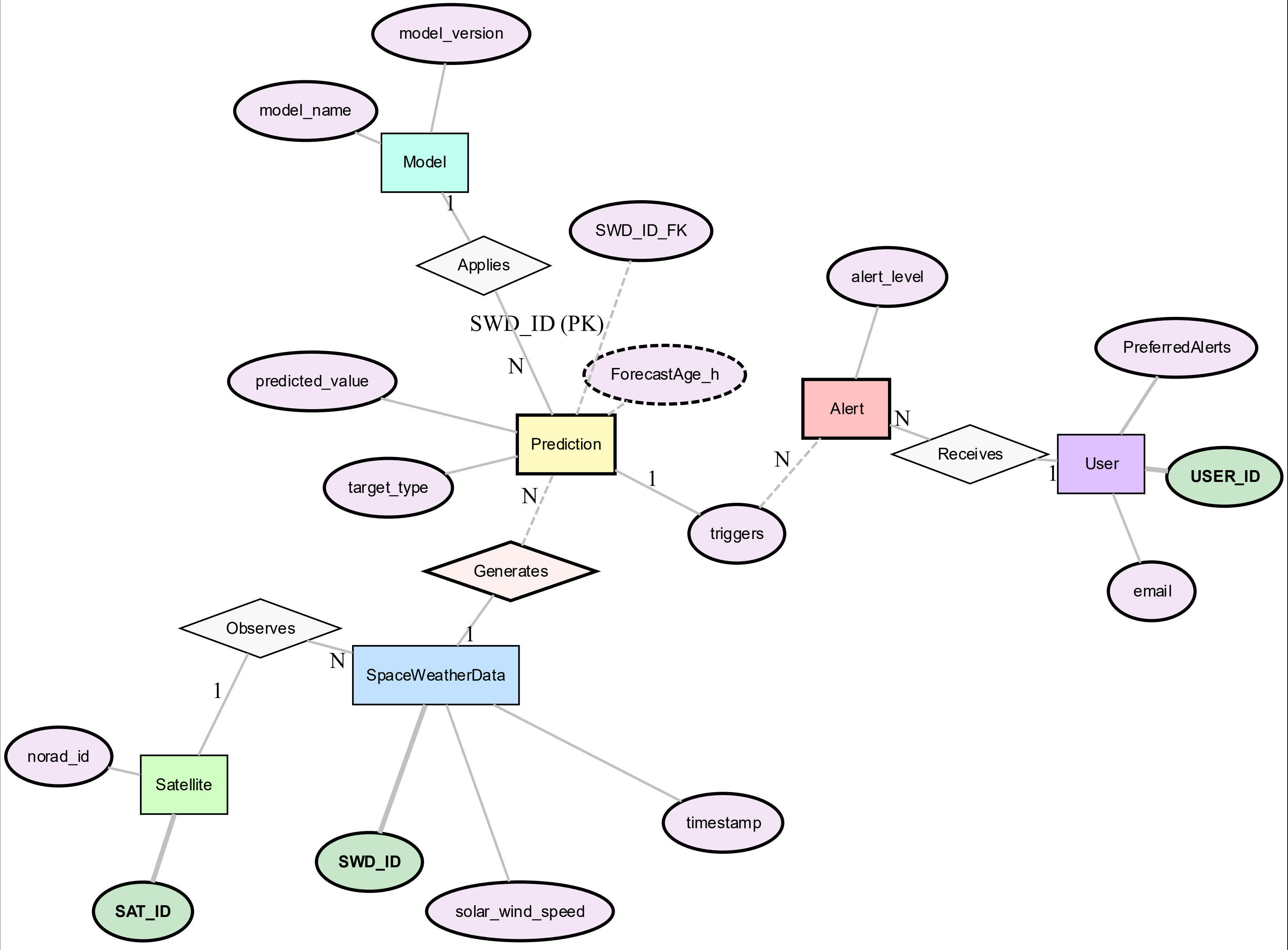
Space Weather Data, Rocket Trajectory, Users, Alerts.

Relationships

- Space Weather Data → Prediction → Alerts
- Rocket Trajectory → User → Alerts

Database

PostgreSQL storing processed and historical data.



CosmoPredict - Chen-Style Entity-Relationship Diagram

UI Mockup: Dashboard

A comprehensive interface for monitoring space weather in real-time.

Real-Time Metrics

Real-time **space weather metrics**: solar wind, proton density, Kp index.

Interactive Charts

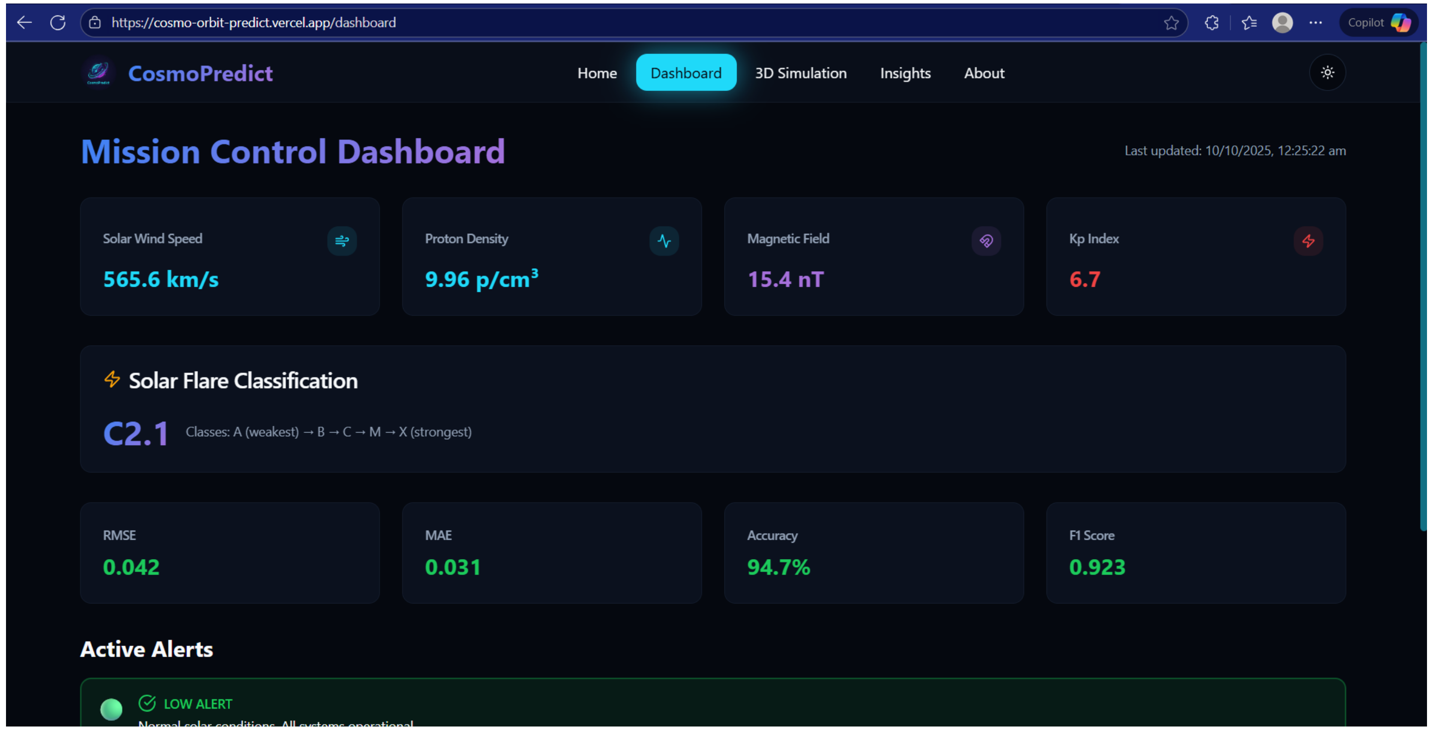
Interactive charts for **historical vs predicted events**.

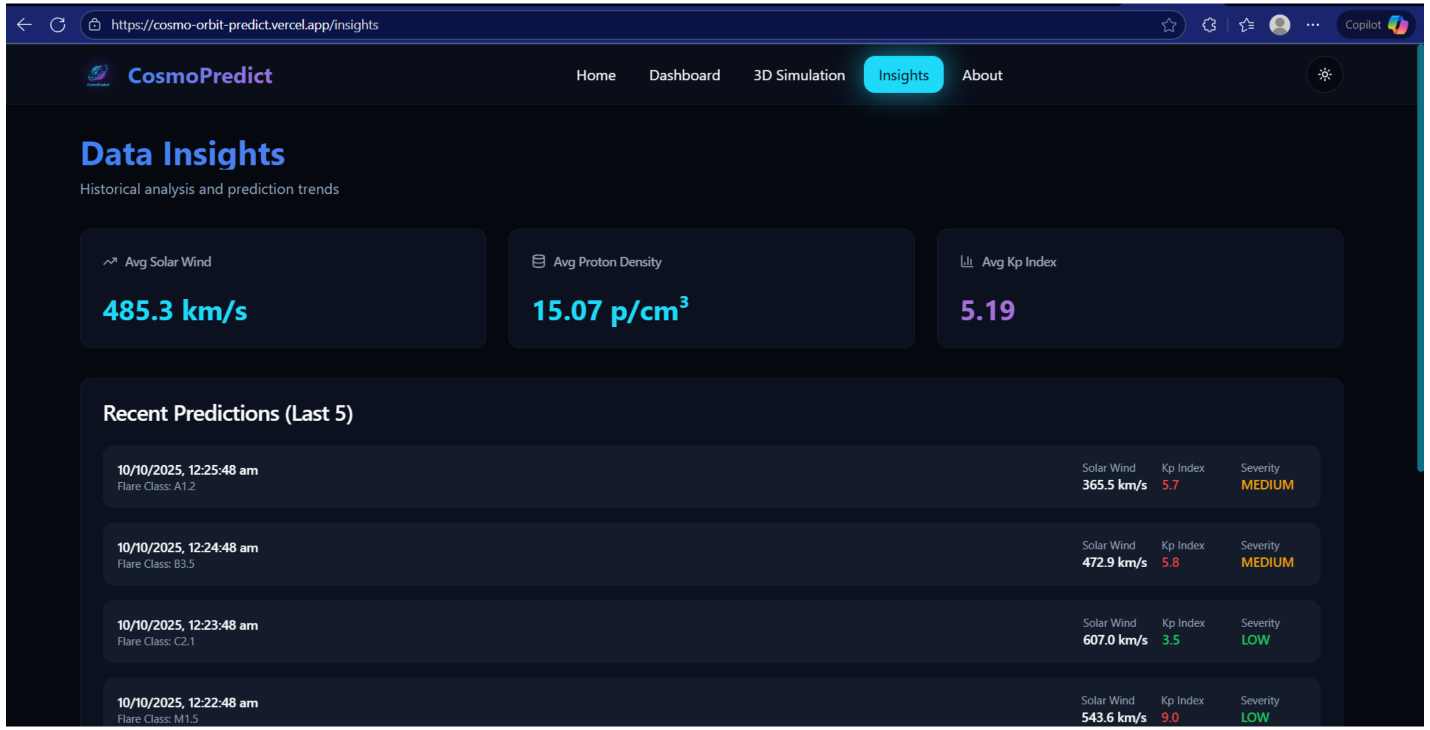
Color-Coded Alerts

Color-coded **alerts** for quick risk assessment.

Theme Toggle

Light/Dark mode toggle for better usability.

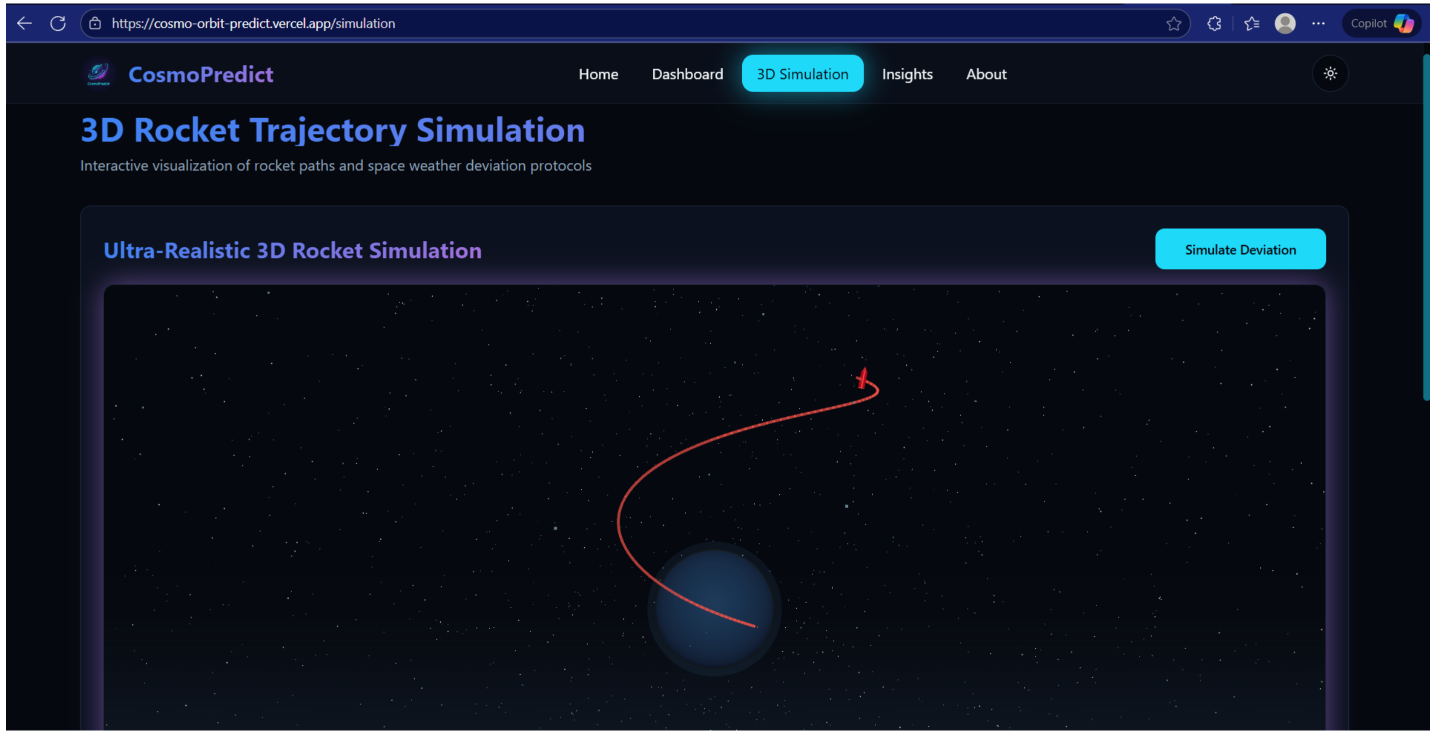


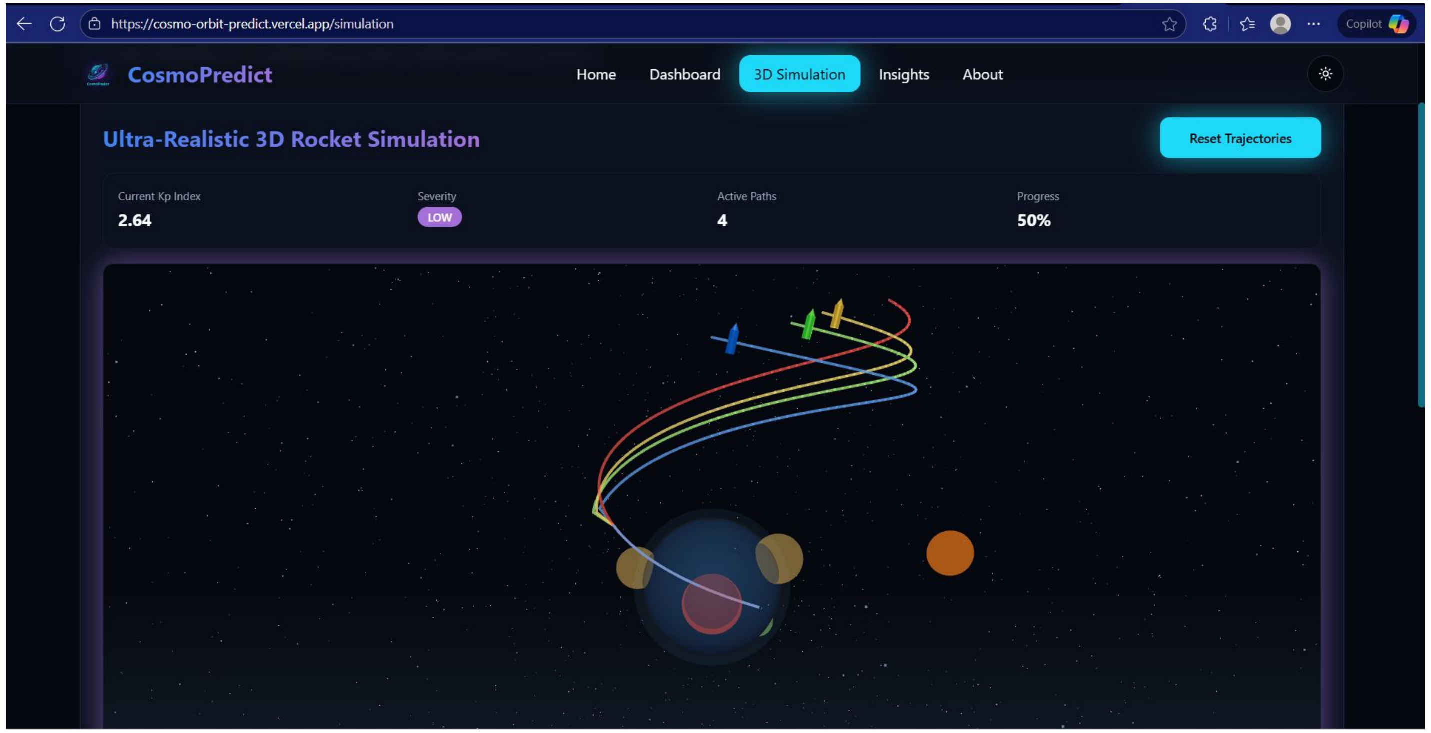


UI Mockup: 3D Rocket Simulation

Interactive Trajectory Visualization

- 3D Earth globe with satellite and rocket trajectories.
- Multiple path simulations: **original trajectory (red), deviation paths (green, yellow, blue)**.
- Highlighted **affected zones** based on predicted space weather.
- User interaction: rotate, zoom, pan, and simulate deviation.





Key Advantages

Proactive Protection

Proactive protection for satellites and astronauts.

Real-Time Monitoring

Centralized real-time monitoring platform.

Predictive ML Models

Predictive ML models for accurate forecasts.

Operational Support

Operational support with orbital deviation suggestions.

Research & Education

Research & education tool for space science enthusiasts.

Future Scope

Multi-Satellite Missions

Extend predictions for multi-satellite missions.

AI Anomaly Detection

Integrate AI anomaly detection for solar wind and geomagnetic data.

Mobile Alerts

Mobile app for real-time alerts on-the-go.

Enhanced 3D Simulation

Enhanced 3D simulation realism with particle effects and trajectory analytics.

Solar Flare Classification

Incorporate solar flare severity classification and risk analytics.

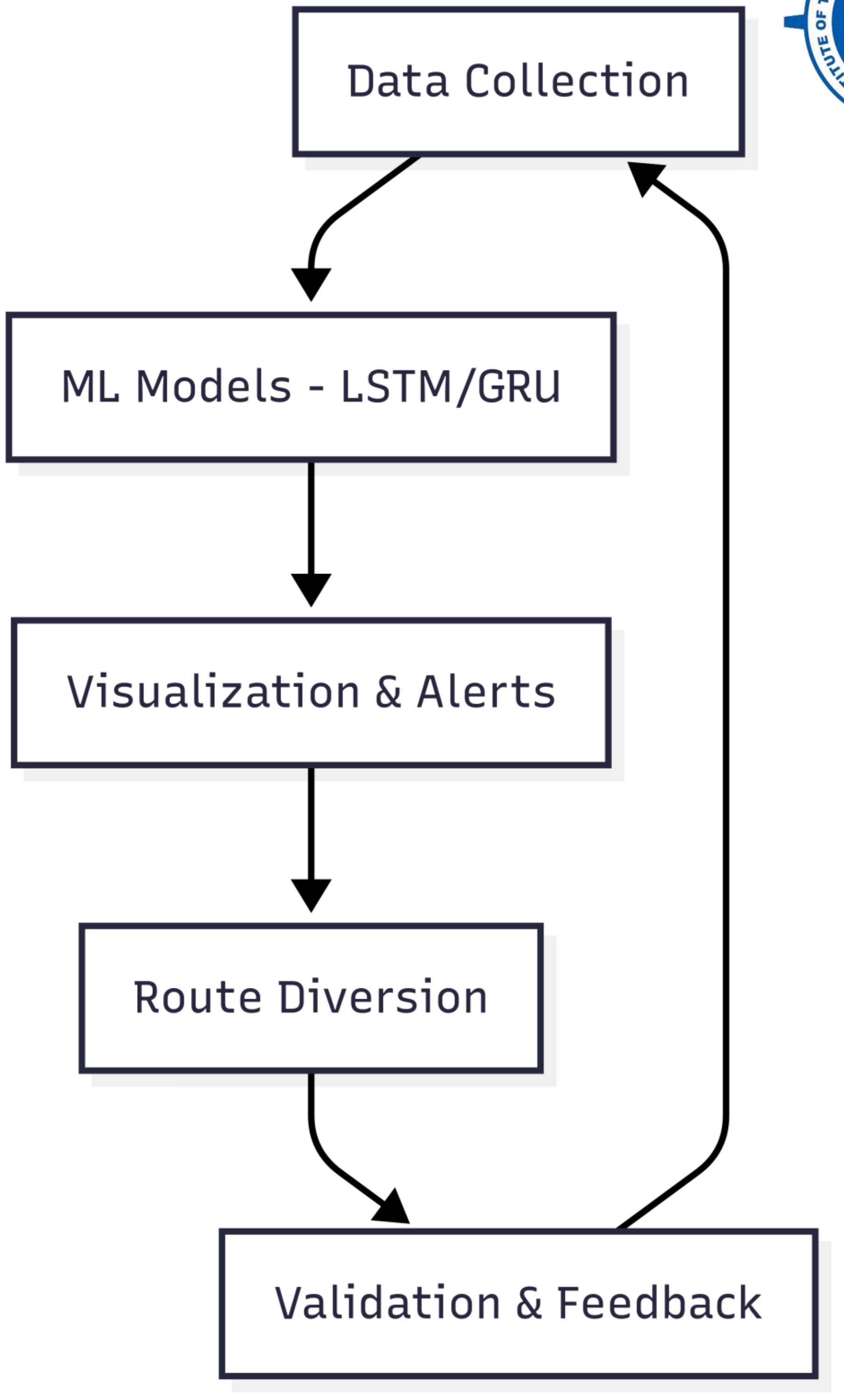
Feasibility & Scalability

Technical Foundation

- Cloud hosting ensures **scalable performance** for multiple users.
- Open-source ML frameworks and APIs reduce **development cost**.

Architecture & Performance

- Modular architecture allows **easy integration** of new data sources.
- Performance metrics ensure **real-time predictions under 5 seconds**.



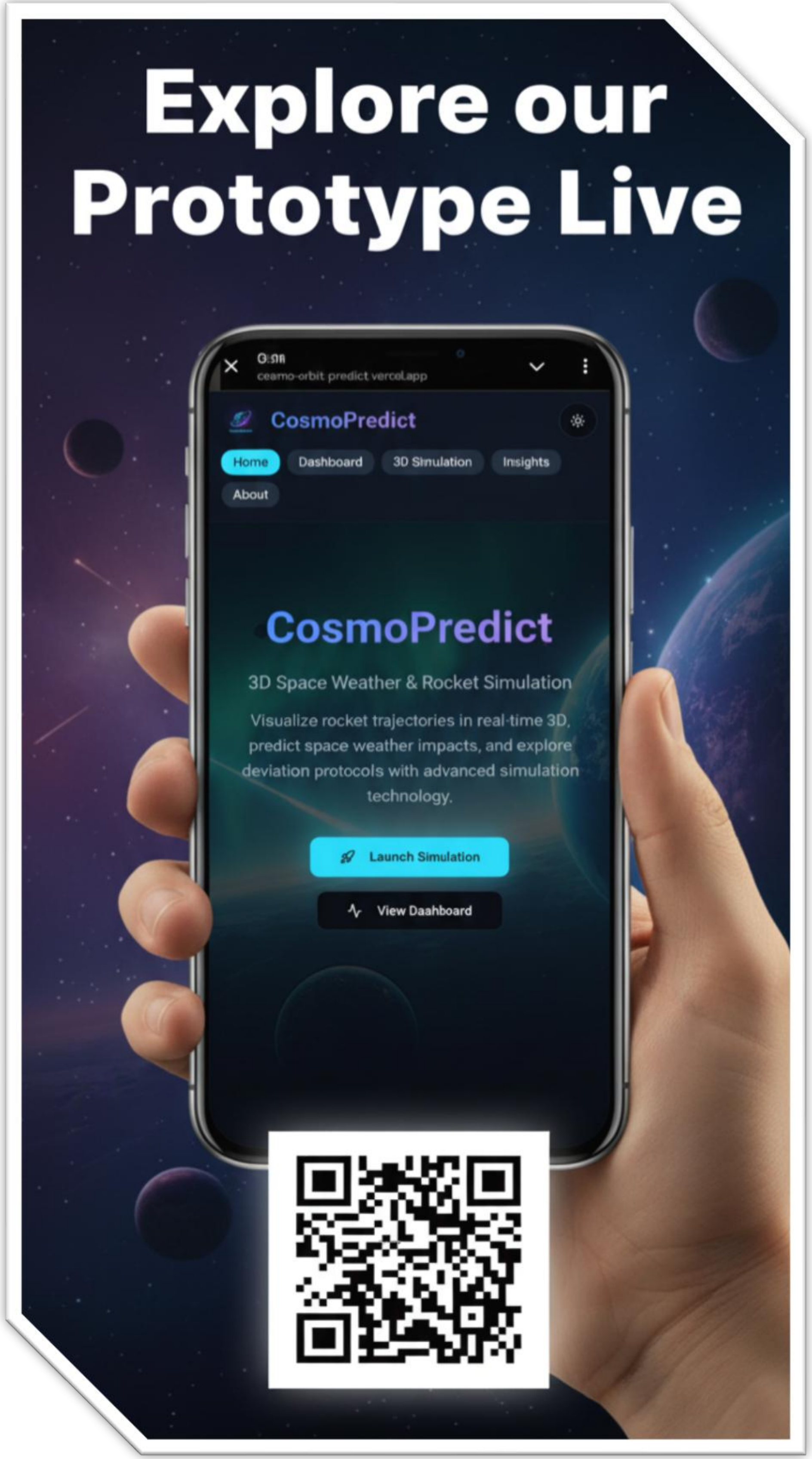
Conclusion

CosmoPredict combines **ML, real-time data, and 3D visualization** for space weather prediction.

Enhances **satellite safety, astronaut protection, and mission planning**.

Acts as a **research and educational tool** for space weather studies.

> "We are confident in completing all tasks successfully and seek approval for project execution."



References

NASA OMNIWeb

https://omniweb.gsfc.nasa.gov

NOAA Space Weather

https://www.swpc.noaa.gov

ACE Satellite Data

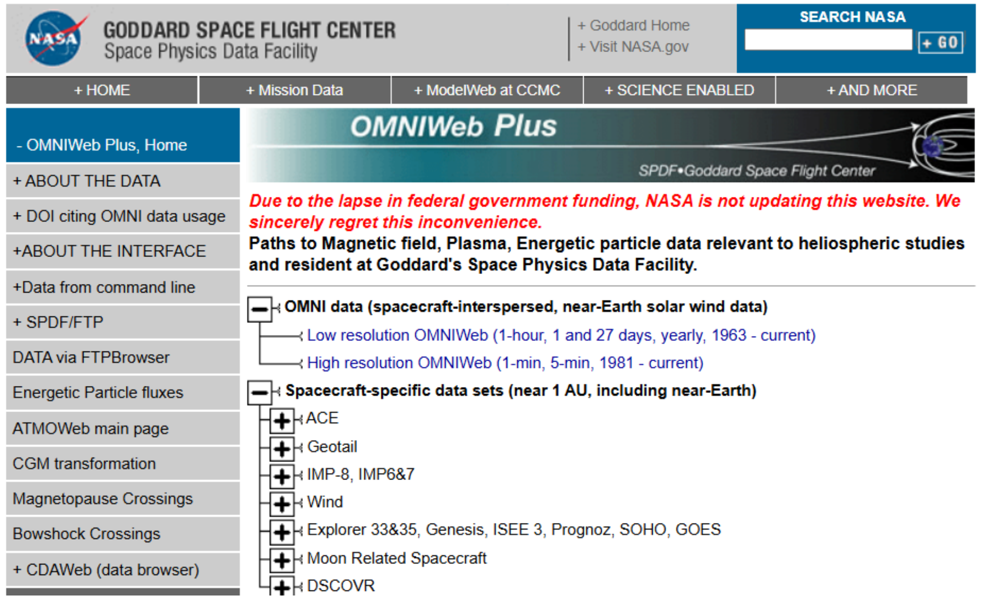
https://www.srl.caltech.edu/ACE/ASC/level2/

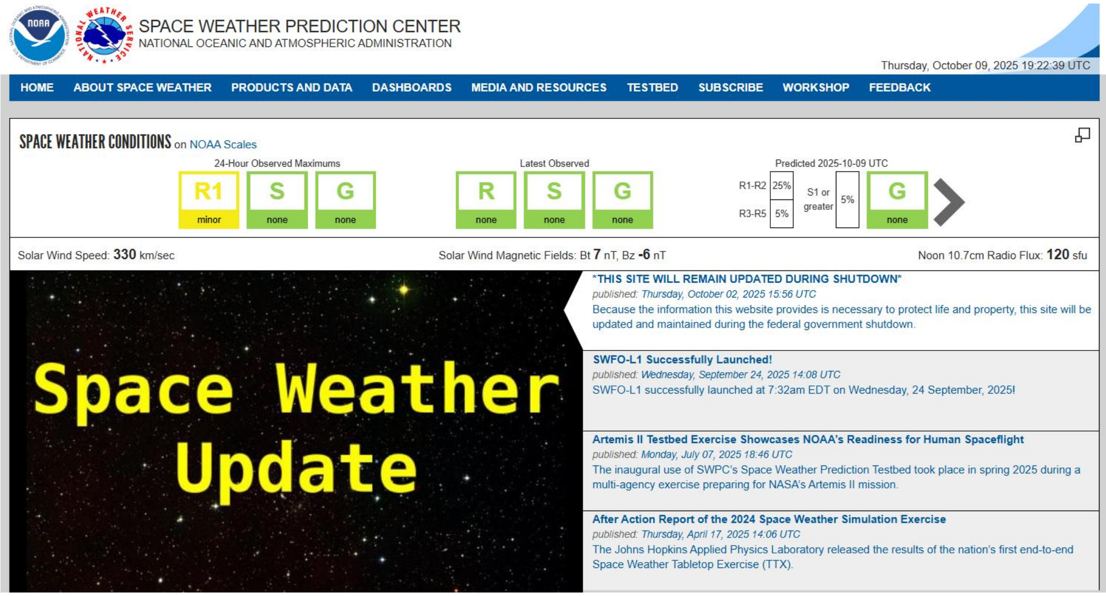
Standards

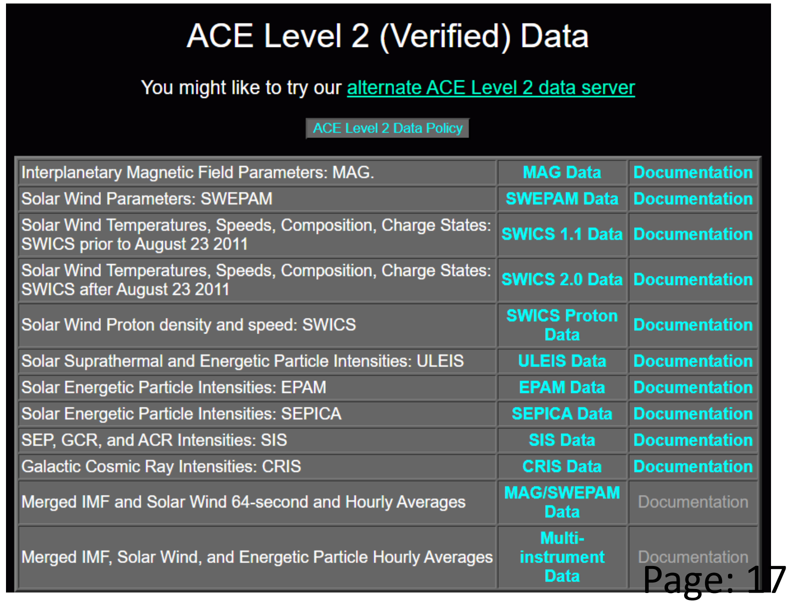
IEEE Std 830-1998 (Software Requirements Specification Standard)

Documentation

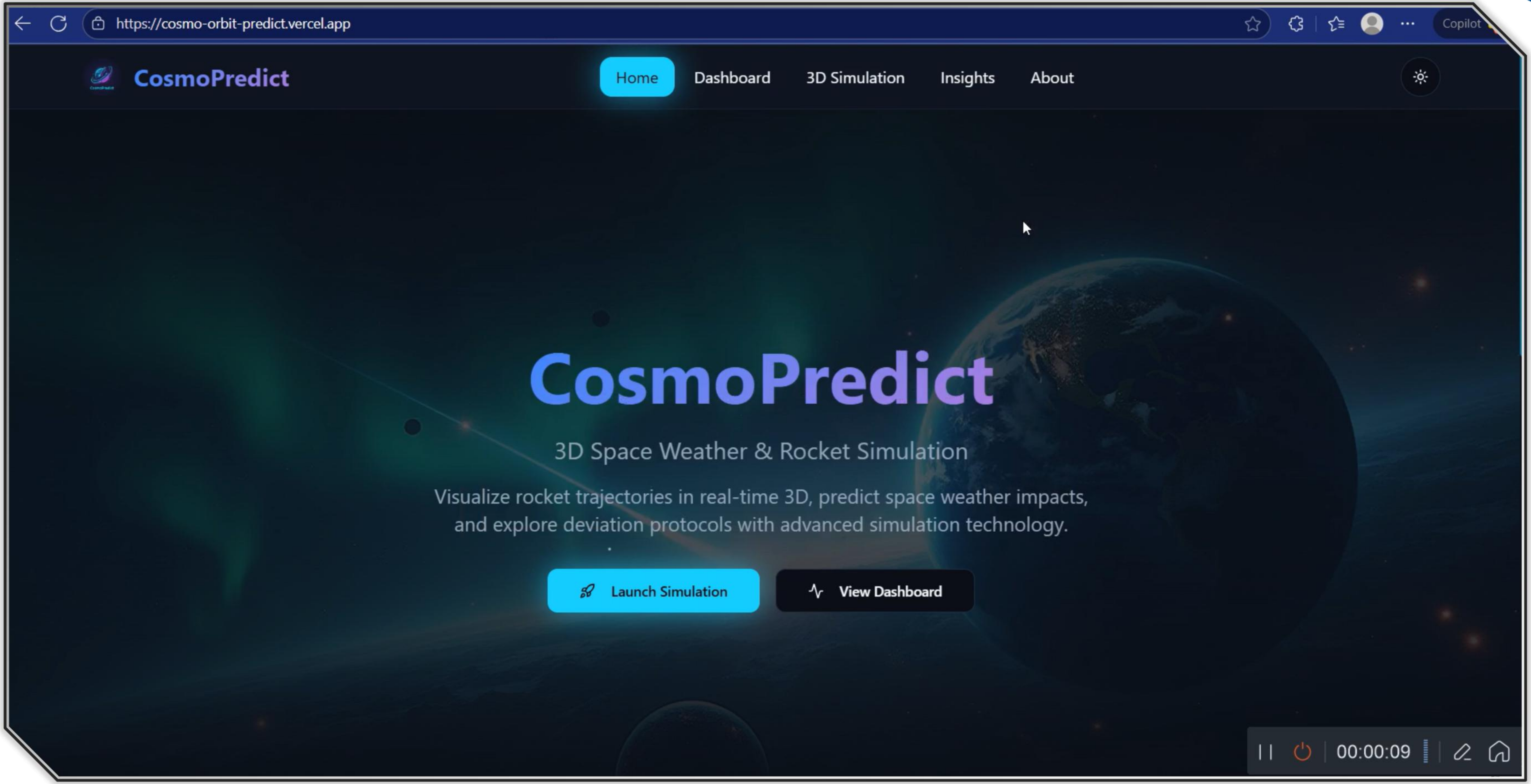
Relevant ML & Python documentation







Demo / Simulation Flow



Impact & Applications

Highlight potential use cases: satellite operators, space agencies, astronaut missions, research labs.

Satellite Operators

Real-time protection and operational guidance for commercial and government satellite fleets.

Space Agencies

Mission planning support and space weather forecasting for exploration programs.

Astronaut Missions

Safety monitoring and radiation exposure predictions for crewed space missions.

Research Labs

Educational platform and research tool for space weather studies and analysis.


THANK YOU